5.6.2 Newton's Laws of Motion

After completing this chapter you should be able to apply models, theories and laws to situations involving energy, force and motion.

For Questions 1 to 12 choose the best answer (A), (B), (C) or (D).

1. Which is an example of Newton's first law of motion?

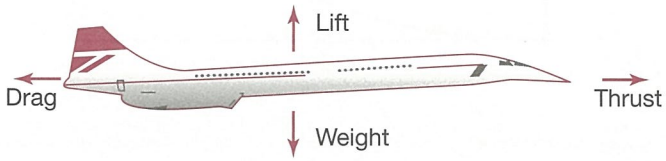
(A) If you slap the desk hard, it hurts your hand.
(B) A bottle dropped on concrete will break, but one dropped onto grass does not break.
(C) The larger the force acting on an object, the faster it goes.
(D) If a bus stops suddenly, you lurch forwards.

2. Which is an example of Newton's third law of motion?

(A) If you catch a hard-hit cricket ball, it hurts your hand.
(B) When the car you are travelling in turns a corner, you lean to the opposite side.
(C) A truck requires more force to accelerate it than a car.
(D) All objects near the Earth's surface fall at the same rate.

Use the following information to answer the next TWO questions.

The diagram shows the forces acting on a plane in flight.



3. The plane is flying at constant speed at a constant altitude. Which statement about these forces is correct?

(A) All four forces must be equal.
(B) Lift will be equal to weight and thrust equal to drag.
(C) Lift will be greater than weight and thrust greater than drag.
(D) Lift and thrust must be equal, both larger than weight or drag.

4. Which statement about the forces acting on the plane as it accelerates and ascends as it travels along the runway during take-off?

(A) Lift will be larger than weight and thrust larger than drag.
(B) Lift will equal weight and thrust will equal drag.
(C) Lift must be greater than weight but thrust will equal drag.
(D) All four forces must be equal.

5. When a car stops suddenly a 60 kg mother and a 15 kg child will both be 'thrown forwards'. Which statement about this is correct?

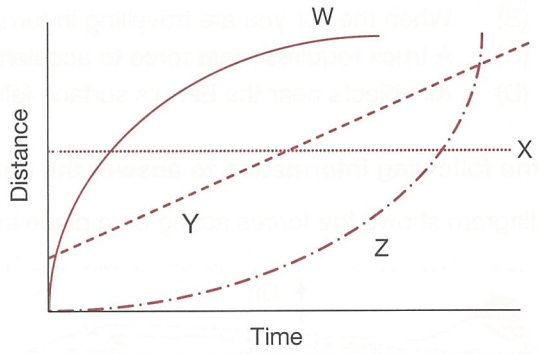
(A) The mother and child will experience the same force because they both stop at the same time.
(B) The mother experiences more force because she has more mass.
(C) The net force on both is zero because the seatbelts hold them in position.
(D) Neither experiences a force. It is their inertia that keeps them moving forwards.

6. Which statement about the relationship between the force acting on a body, the mass of the body and its acceleration is correct?

(A) The larger the force, the larger the acceleration regardless of the mass of the body.
(B) The larger the mass of the body, the less the acceleration produced by a constant force.
(C) The larger the mass, the larger the force needed to accelerate it.
(D) The same force will produce the same acceleration regardless of the mass of the body.

Use the following information to answer the next THREE questions.

The diagram shows a graph of the motion of four objects.



7. Which object is moving with constant speed?

(A) W
(B) X
(C) Y
(D) Z

8. Which object is stationary?

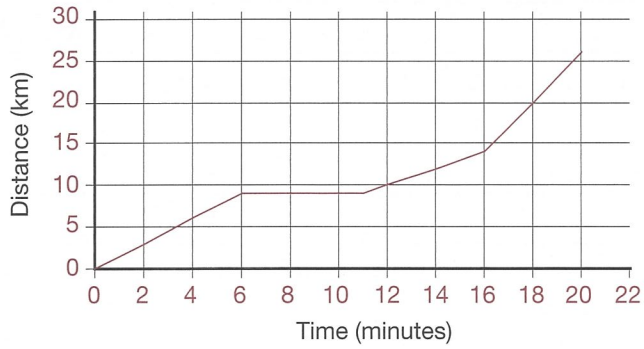
(A) W
(B) X
(C) Y
(D) Z

9. Which object is slowing down?

(A) W
(B) X
(C) Y
(D) Z

Use the following information to answer the next THREE questions.

The graph represents Tina's journey as she rode her bicycle from a friend's place back to her home.



10. How far is it from Tina's friend's place to home?

(A) 9 km.
(B) 20 km.
(C) 25 km.
(D) 26 km.

11. Tina stopped for a rest on the way home. For how long did she stop?

(A) 4 minutes.
(B) 5 minutes.
(C) 9 minutes.
(D) 11 minutes.

12. Between which times was Tina moving fastest?

(A) Between 0 and 6 minutes.
(B) Between 6 and 11 minutes.
(C) Between 11 and 16 minutes.
(D) Between 16 and 20 minutes.

13. Recall Newton's laws of motion:

Law 1

An object at rest or moving with constant motion will continue to do so unless an unbalanced force acts on it.

Law 2

The acceleration produced when an unbalanced force acts on an object is directly proportional to the magnitude of the force applied.

Law 3

for every action there is an equal and opposite reaction.

14. You are trying to explain these laws to someone. Give an example to illustrate each of these laws of motion that you could use to help your explanation.

Law 1

A book sitting on a table will not slide across the table unless pushed.

Law 2

The harder the book is pushed, the faster it will move across the table.

Law 3

when the book is sitting on the table, the weight of the table is supported by the reaction force of the table acting upwards.

15. What is meant by an unbalanced force?

One that will cause an object to move with an acceleration.

16. The ticker timer tapes below show the record of motion of four objects.



(a) Which tapes indicate the objects moving with constant speed?

A and C

(b) In your own words, describe the motion of the objects indicated by the other tapes.

B - accelerating

D - accelerating then decelerating.

17. The table shows the results of an experiment done to study the effect of different forces acting on various trolleys. Use these results to answer the questions that follow them.

Mass of trolley (kg)	Applied force (N)	Acceleration produced ($m\ s^{-2}$)
1.5	1.0	0.67
1.5	2.0	1.33
1.5	3.0	2.00
1.5	4.0	2.67

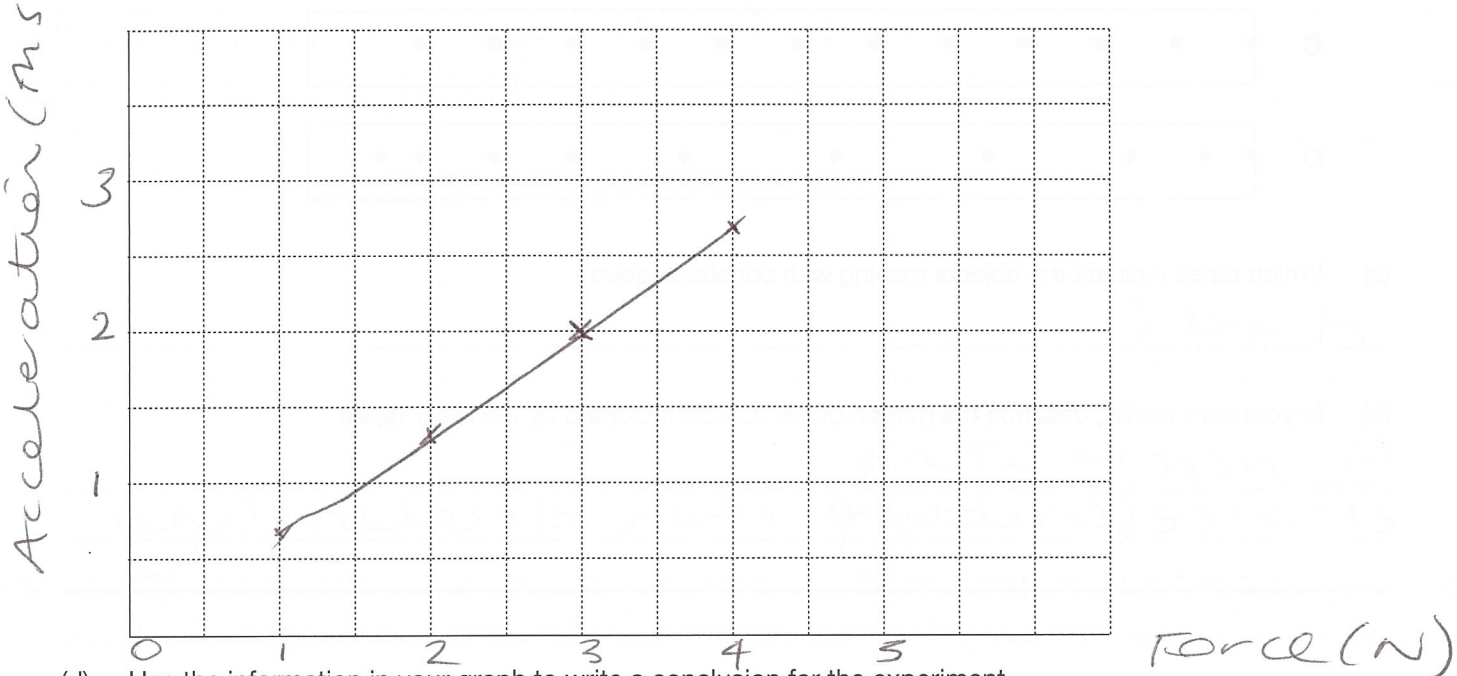
(a) What is the aim of the experiment?

To examine the relationship between the force applied to an object of constant mass and the acceleration produced.

(b) Complete the table:

Independent variable	Dependent variable	Controlled variable
Force applied	Acceleration of trolley	mass of the trolley

(c) Graph the results of the experiment on the grid below.



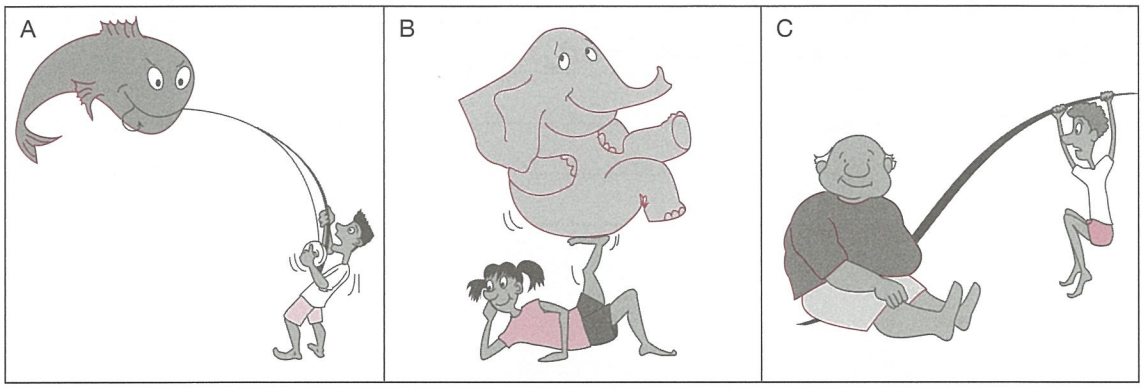
(d) Use the information in your graph to write a conclusion for the experiment.

The acceleration of the trolley was directly proportional to the force applied

(e) Suggest two ways the experiment could be improved.

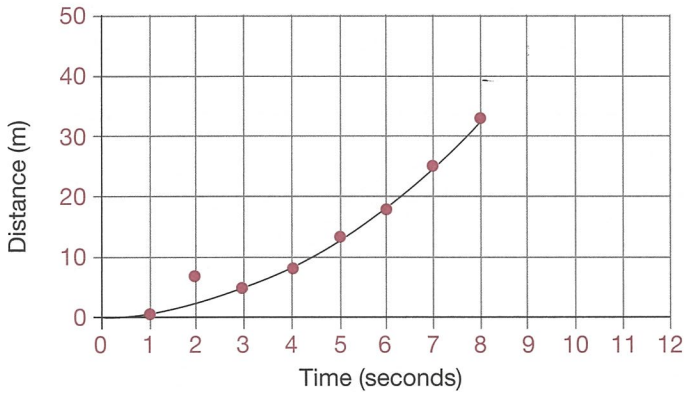
Repeats.
Repeat with different masses.

18. Label and describe the action-reaction force pairs in each diagram. Indicate clearly which object the forces are acting on (two pairs in A and C).



	Action force	Reaction force
A	Fish pulling line out.	Line pulling fish in
B	weight of elephant on foot.	Little's foot pushing up on elephant
C	weight of person pulling on bar	Tension of bar pulling up on person.

19. The graph shows the results of an experiment in which a student recorded the position of a moving trolley each second for 8 seconds.



(a) Describe the motion of the trolley during the 8 seconds.

Accelerates

(b) What distance had the trolley covered at time 7 seconds?

~25m

(c) Estimate how long it would take the trolley to travel 45 m if its motion continued in the same pattern.

~ 9 seconds

(d) The plot point at 2.0 seconds has been ignored when the graph was drawn. Is this omission justified? Explain your answer.

Yes. it doesnt fit the overall trend (is an outlier).